

**Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark.**

| dataset            | feature                    | accuracy           | precision          | recall             | f1                 | mutual_information |
|--------------------|----------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| phiusiil           | URLSimilarityIndex         | 0.9967374960991802 | 1.0                | 0.992344051661008  | 0.9961573161359308 | 0.662215786810744  |
| phiusiil           | NoOfExternalRef            | 0.9611336491815372 | 0.9394193008433658 | 0.9714399840223686 | 0.955161353668914  | 0.5654485944505745 |
| phiusiil           | LineOfCode                 | 0.9609350619875742 | 0.9653478854024556 | 0.9421476599427469 | 0.9536066844109026 | 0.595168316462994  |
| phiusiil           | NoOfSelfRef                | 0.948707764759284  | 0.9019774870702768 | 0.9868850276279876 | 0.9425228891149542 | 0.5293939297844878 |
| phiusiil           | NoOfImage                  | 0.9421260177593692 | 0.9423430791249232 | 0.920511284202117  | 0.9312992523742172 | 0.5494509954646463 |
| phiusiil           | NoOfJS                     | 0.9319697012681212 | 0.9302610948258232 | 0.9084614872511816 | 0.9192320646682384 | 0.504826032305074  |
| phiusiil           | NoOfCSS                    | 0.9018695565831656 | 0.9678698607963744 | 0.7961520537913588 | 0.8736530664426343 | 0.4556611168670177 |
| phiusiil           | HasSocialNet               | 0.8826633379670346 | 0.7863720073664825 | 0.9949404167498836 | 0.8784458943161112 | 0.3911324693936928 |
| phiusiil           | HasCopyrightInfo           | 0.8674288632301626 | 0.7883736484226953 | 0.9416816456960256 | 0.8582349907472014 | 0.3219704649717501 |
| phiusiil           | LargestLineLength          | 0.8384635025107096 | 0.9557314570507184 | 0.6510884761334133 | 0.7745307674031836 | 0.4585741629380953 |
| phiusiil           | HasDescription             | 0.8328463218814718 | 0.7326808380486313 | 0.956860395446375  | 0.8298978001039321 | 0.2806251321292377 |
| phiusiil           | NoOfOtherSpecialCharsInURL | 0.793838123067321  | 0.9352267624607096 | 0.554623527062113  | 0.6963099168373104 | 0.2347809746872036 |
| phiusiil           | IsHTTPS                    | 0.791568555136316  | 1.0                | 0.5108847613341322 | 0.6762723066754792 | 0.2348072550255246 |
| phiusiil           | DomainTitleMatchScore      | 0.789866379188062  | 0.7191205249222977 | 0.8317688569336262 | 0.7713536039512271 | 0.1912685189780731 |
| phiusiil           | IsResponsive               | 0.7814122386450679 | 0.7770372614359286 | 0.683043738765728  | 0.7270150575730735 | 0.1665703678319596 |
| uci                | sslfinal_state             | 0.8735763097949886 | 0.90521327014218   | 0.8432671081677704 | 0.8731428571428571 | 0.3447768589996409 |
| uci                | url_of_anchor              | 0.8154897494305239 | 0.9965870307167236 | 0.6445916114790287 | 0.7828418230563002 | 0.3104458656375795 |
| uci                | having_sub_domain          | 0.6845102505694761 | 0.6506849315068494 | 0.8388520971302428 | 0.7328833172613307 | 0.0806267546036059 |
| uci                | web_traffic                | 0.6674259681093394 | 0.6709129511677282 | 0.6975717439293598 | 0.683982683982684  | 0.058522520357888  |
| uci                | prefix_suffix              | 0.6378132118451025 | 0.5875486381322957 | 1.0                | 0.7401960784313726 | 0.104921861669007  |
| uci                | sfh                        | 0.6150341685649203 | 0.58887171561051   | 0.8410596026490066 | 0.6927272727272727 | 0.0462648656175914 |
| uci                | links_in_tags              | 0.5968109339407744 | 0.6460176991150443 | 0.4834437086092715 | 0.553030303030303  | 0.0393409056227036 |
| uci                | request_url                | 0.591116173120729  | 0.6236842105263158 | 0.5231788079470199 | 0.5690276110444178 | 0.032647679112516  |
| uci                | domain_registration_length | 0.5842824601366743 | 0.6506849315068494 | 0.4194260485651214 | 0.5100671140939598 | 0.0179794658834226 |
| uci                | having_ip_address          | 0.571753986332574  | 0.5955334987593052 | 0.5298013245033113 | 0.5607476635514018 | 0.0                |
| uci                | links_pointing_to_page     | 0.5637813211845103 | 0.5708502024291497 | 0.6225165562913907 | 0.5955649419218585 | 0.0055553214272923 |
| uci                | google_index               | 0.5318906605922551 | 0.63125            | 0.2229580573951435 | 0.3295269168026101 | 0.0                |
| uci                | age_of_domain              | 0.5296127562642369 | 0.5444444444444444 | 0.5408388520971302 | 0.5426356589147286 | 0.0090995515578689 |
| uci                | abnormal_url               | 0.5296127562642369 | 0.5268817204301075 | 0.8653421633554084 | 0.6549707602339181 | 0.0129948777383075 |
| uci                | statistical_report         | 0.5250569476082004 | 0.6363636363636364 | 0.1854304635761589 | 0.2871794871794871 | 0.0                |
| mendeley_hannousse | google_index               | 0.8548104956268222 |                    |                    | 0.8611266034578918 | 0.2994122171059308 |
| mendeley_hannousse | web_traffic                | 0.7877551020408163 |                    |                    | 0.7863849765258216 | 0.2984626354884387 |
| mendeley_hannousse | nb_hyperlinks              | 0.7772594752186589 |                    |                    | 0.7983104540654699 | 0.2336224271865306 |
| mendeley_hannousse | page_rank                  | 0.7714285714285715 |                    |                    | 0.7615571776155717 | 0.2211787638339284 |
| mendeley_hannousse | ratio_intHyperlinks        | 0.7288629737609329 |                    |                    | 0.6978557504873294 | 0.2444913886895223 |
| mendeley_hannousse | nb_www                     | 0.7276967930029155 |                    |                    | 0.7426997245179063 | 0.1107767520171292 |
| mendeley_hannousse | ratio_extHyperlinks        | 0.7247813411078717 |                    |                    | 0.7114914425427873 | 0.2402741493444735 |
| mendeley_hannousse | safe_anchor                | 0.7201166180758017 |                    |                    | 0.7306397306397306 | 0.1792403962425659 |

**Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark. (continued)**

|                                             |                                    |                    |                    |                    |
|---------------------------------------------|------------------------------------|--------------------|--------------------|--------------------|
| mendeley_hannousse                          | longest_word_path                  | 0.6973760932944606 | 0.6670942912123156 | 0.1199855314109274 |
| mendeley_hannousse                          | links_in_tags                      | 0.6903790087463557 | 0.7093596059113301 | 0.1209079961488006 |
| mendeley_hannousse                          | ratio_extRedirection               | 0.6892128279883382 | 0.7003934794828556 | 0.1208953249578295 |
| mendeley_hannousse                          | domain_age                         | 0.6880466472303207 | 0.6555054732775274 | 0.279647785006152  |
| mendeley_hannousse                          | shortest_word_host                 | 0.6810495626822157 | 0.6806771745475774 | 0.0738478244125739 |
| mendeley_hannousse                          | char_repeat                        | 0.6740524781341107 | 0.680022896393818  | 0.08819377638949   |
| mendeley_hannousse                          | ratio_digits_url                   | 0.673469387755102  | 0.6534653465346535 | 0.1020362646838448 |
| mendeley_tan                                | PctExtHyperlinks                   | 0.8886666666666667 | 0.8876933422999328 | 0.4694645544898392 |
| mendeley_tan                                | PctExtResourceUrls                 | 0.7793333333333333 | 0.7535368577810871 | 0.2938791675531358 |
| mendeley_tan                                | PctExtNullSelfRedirectHyperlinksRT | 0.7793333333333333 | 0.7257663628831814 | 0.2166061331654218 |
| mendeley_tan                                | PctNullSelfRedirectHyperlinks      | 0.7353333333333333 | 0.7800554016620499 | 0.2341867873312832 |
| mendeley_tan                                | ExtMetaScriptLinkRT                | 0.7306666666666667 | 0.7548543689320388 | 0.1217853502396473 |
| mendeley_tan                                | NumNumericChars                    | 0.7226666666666667 | 0.7409713574097135 | 0.1245147404157263 |
| mendeley_tan                                | FrequentDomainNameMismatch         | 0.6986666666666667 | 0.5830258302583026 | 0.1318419695965218 |
| mendeley_tan                                | NumDash                            | 0.6693333333333333 | 0.7259668508287292 | 0.1132495764307262 |
| mendeley_tan                                | SubmitInfoToEmail                  | 0.6346666666666667 | 0.7295162882527148 | 0.077231458698451  |
| mendeley_tan                                | InsecureForms                      | 0.6246666666666667 | 0.7189216175736396 | 0.0586641863473267 |
| mendeley_tan                                | PathLevel                          | 0.6233333333333333 | 0.6686217008797654 | 0.0628910137177032 |
| mendeley_tan                                | IframeOrFrame                      | 0.62               | 0.6750285062713797 | 0.0355050630909328 |
| mendeley_tan                                | UrlLength                          | 0.6173333333333333 | 0.6550480769230769 | 0.0419254325406448 |
| mendeley_tan                                | NumDots                            | 0.6153333333333333 | 0.5612167300380229 | 0.0611992759683925 |
| mendeley_tan                                | PathLength                         | 0.604              | 0.6925465838509317 | 0.0546984906819405 |
| uci379_website_phishing                     | SFH                                | 0.8325123152709359 | 0.8583333333333333 | 0.2562112801883558 |
| uci379_website_phishing                     | SSLfinal_State                     | 0.7881773399014779 | 0.8138528138528138 | 0.1673532319129793 |
| uci379_website_phishing                     | popUpWindow                        | 0.7438423645320197 | 0.7886178861788617 | 0.1469565851250001 |
| uci379_website_phishing                     | URL_Length                         | 0.6157635467980296 | 0.7045454545454546 | 0.0175734580203541 |
| uci379_website_phishing                     | web_traffic                        | 0.6059113300492611 | 0.6923076923076923 | 0.0371275650453013 |
| uci379_website_phishing                     | having_IP_Address                  | 0.5960591133004927 | 0.7469135802469136 | 0.0                |
| uci379_website_phishing                     | Request_URL                        | 0.5911330049261084 | 0.6556016597510373 | 0.0467234326648795 |
| uci379_website_phishing                     | age_of_domain                      | 0.5862068965517241 | 0.6585365853658537 | 0.0                |
| uci379_website_phishing                     | URL_of_Anchor                      | 0.5763546798029556 | 0.6293103448275862 | 0.0406746833848628 |
| uci379_website_phishing__suspicious_dropped | SFH                                | 0.8776595744680851 | 0.8959276018099548 | 0.347106157184478  |
| uci379_website_phishing__suspicious_dropped | SSLfinal_State                     | 0.8351063829787234 | 0.8544600938967136 | 0.158212415188756  |
| uci379_website_phishing__suspicious_dropped | popUpWindow                        | 0.75               | 0.7873303167420814 | 0.1731931852559332 |
| uci379_website_phishing__suspicious_dropped | URL_of_Anchor                      | 0.6276595744680851 | 0.6788990825688074 | 0.0638068842395207 |
| uci379_website_phishing__suspicious_dropped | Request_URL                        | 0.5957446808510638 | 0.5064935064935064 | 0.0496499056762813 |

**Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark. (continued)**

|                                                 |                   |                    |                    |                    |
|-------------------------------------------------|-------------------|--------------------|--------------------|--------------------|
| uci379_website_phishing__sus<br>picious_dropped | URL_Length        | 0.5797872340425532 | 0.6609442060085837 | 0.0105902350567534 |
| uci379_website_phishing__sus<br>picious_dropped | web_traffic       | 0.5691489361702128 | 0.6462882096069869 | 0.0622995873948881 |
| uci379_website_phishing__sus<br>picious_dropped | age_of_domain     | 0.5691489361702128 | 0.6301369863013698 | 0.0527353726561756 |
| uci379_website_phishing__sus<br>picious_dropped | having_IP_Address | 0.5638297872340425 | 0.7210884353741497 | 0.0                |

**Table T4. Progressive removal on PhiUSIL: held-out performance after greedily removing the highest-attributed feature at each iteration.**

```
[record 1 of 15]
  iteration : 0
  n_features_remaining : 50
  removed_feature : URLSimilarityIndex
  accuracy : 1.0
  f1 : 1.0
  mcc : 1.0
  top_feature_after : LineOfCode
  removed_so_far : 0
removed_features_so_far :
  top_feature_before : URLSimilarityIndex
  top_shap_before : 0.247806890352753
  second_feature : LineOfCode
  second_shap : 0.1023126702474286
  fit_seconds : 21.82
  shap_seconds : 0.01
hyperparameter_regime : retuned_per_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                 therefore understate their feature set

[record 2 of 15]
  iteration : 1
  n_features_remaining : 49
  removed_feature : LineOfCode
  accuracy : 0.9997163040086244
  f1 : 0.9996671105193076
  mcc : 0.9994199473420988
  top_feature_after : IsHTTPS
  removed_so_far : 1
removed_features_so_far : URLSimilarityIndex
  top_feature_before : LineOfCode
  top_shap_before : 0.1152708303479257
  second_feature : IsHTTPS
  second_shap : 0.109668183553169
  fit_seconds : 30.77
  shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                 therefore understate their feature set

[record 3 of 15]
  iteration : 2
  n_features_remaining : 48
  removed_feature : IsHTTPS
  accuracy : 0.9995460864137988
  f1 : 0.9994672704268496
  mcc : 0.9990719648501164
  top_feature_after : NoOfSelfRef
  removed_so_far : 2
removed_features_so_far : URLSimilarityIndex; LineOfCode
  top_feature_before : IsHTTPS
  top_shap_before : 0.126553705349837
  second_feature : NoOfSelfRef
  second_shap : 0.0625032237839399
  fit_seconds : 28.21
  shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                 therefore understate their feature set

[record 4 of 15]
  iteration : 3
```

**Table T4. Progressive removal on PhiUSIL: held-out performance after greedily removing the highest-attributed feature at each iteration. (continued)**

```

n_features_remaining : 47
  removed_feature : NoOfSelfRef
    accuracy : 0.9981559760560584
    f1 : 0.9978342718155466
    mcc : 0.9962301516836702
  top_feature_after : NoOfExternalRef
  removed_so_far : 3
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS
top_feature_before : NoOfSelfRef
top_shap_before : 0.1063587436788972
second_feature : NoOfImage
second_shap : 0.0736897472085234
fit_seconds : 29.99
shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                 therefore understate their feature set

[record 5 of 15]
  iteration : 4
n_features_remaining : 46
  removed_feature : NoOfExternalRef
    accuracy : 0.9978439104655452
    f1 : 0.9974661598986464
    mcc : 0.9955936724655122
  top_feature_after : NoOfImage
  removed_so_far : 4
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef
top_feature_before : NoOfExternalRef
top_shap_before : 0.0922275283717792
second_feature : NoOfImage
second_shap : 0.0850857766769547
fit_seconds : 30.4
shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                 therefore understate their feature set

[record 6 of 15]
  iteration : 5
n_features_remaining : 45
  removed_feature : NoOfImage
    accuracy : 0.9977304320689948
    f1 : 0.997332977730364
    mcc : 0.995361320722614
  top_feature_after : HasSocialNet
  removed_so_far : 5
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef
top_feature_before : NoOfImage
top_shap_before : 0.1030192177679219
second_feature : HasSocialNet
second_shap : 0.0522758004645366
fit_seconds : 30.87
shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                 therefore understate their feature set

[record 7 of 15]
  iteration : 6
n_features_remaining : 44
  removed_feature : HasSocialNet

```

**Table T4. Progressive removal on PhiUSIIL: held-out performance after greedily removing the highest-attributed feature at each iteration. (continued)**

```

accuracy : 0.9973616272802064
f1 : 0.9968986560843032
mcc : 0.9946080630085278
top_feature_after : NoOfJS
removed_so_far : 6
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage
top_feature_before : HasSocialNet
top_shap_before : 0.0748647143718659
second_feature : NoOfJS
second_shap : 0.0727016331505891
fit_seconds : 30.32
shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                therefore understate their feature set

[record 8 of 15]
iteration : 7
n_features_remaining : 43
removed_feature : NoOfJS
accuracy : 0.9969928224914182
f1 : 0.9964654884961652
mcc : 0.9938532601136494
top_feature_after : NoOfCSS
removed_so_far : 7
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
                        HasSocialNet
top_feature_before : NoOfJS
top_shap_before : 0.0854280947188425
second_feature : NoOfOtherSpecialCharsInURL
second_shap : 0.0511721463435716
fit_seconds : 32.01
shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                therefore understate their feature set

[record 9 of 15]
iteration : 8
n_features_remaining : 42
removed_feature : NoOfCSS
accuracy : 0.9965672785043548
f1 : 0.9959622251142924
mcc : 0.9929862970428124
top_feature_after : LargestLineLength
removed_so_far : 8
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
                        HasSocialNet; NoOfJS
top_feature_before : NoOfCSS
top_shap_before : 0.0897294430616568
second_feature : NoOfOtherSpecialCharsInURL
second_shap : 0.0529015924619612
fit_seconds : 27.76
shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                therefore understate their feature set

[record 10 of 15]
iteration : 9
n_features_remaining : 41
removed_feature : LargestLineLength

```

**Table T4. Progressive removal on PhiUSIIL: held-out performance after greedily removing the highest-attributed feature at each iteration. (continued)**

```

        accuracy : 0.995375755340577
        f1 : 0.9945610464146284
        mcc : 0.9905481677112188
    top_feature_after : LetterRatioInURL
    removed_so_far : 9
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
                        HasSocialNet; NoOfJS; NoOfCSS
    top_feature_before : LargestLineLength
    top_shap_before : 0.0607343714721147
    second_feature : LetterRatioInURL
    second_shap : 0.0535198590666558
    fit_seconds : 23.92
    shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                  therefore understate their feature set

[record 11 of 15]
    iteration : 10
    n_features_remaining : 40
    removed_feature : LetterRatioInURL
        accuracy : 0.9945814065647252
        f1 : 0.9936237689868136
        mcc : 0.9889254016453218
    top_feature_after : NoOfOtherSpecialCharsInURL
    removed_so_far : 10
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
                        HasSocialNet; NoOfJS; NoOfCSS; LargestLineLength
    top_feature_before : LetterRatioInURL
    top_shap_before : 0.0583808986068427
    second_feature : NoOfOtherSpecialCharsInURL
    second_shap : 0.054602624538909
    fit_seconds : 23.86
    shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                  therefore understate their feature set

[record 12 of 15]
    iteration : 11
    n_features_remaining : 39
    removed_feature : NoOfOtherSpecialCharsInURL
        accuracy : 0.9927657522199212
        f1 : 0.9914826814522864
        mcc : 0.9852132882490036
    top_feature_after : NoOfLettersInURL
    removed_so_far : 11
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
                        HasSocialNet; NoOfJS; NoOfCSS; LargestLineLength; LetterRatioInURL
    top_feature_before : NoOfOtherSpecialCharsInURL
    top_shap_before : 0.0654809384262721
    second_feature : HasCopyrightInfo
    second_shap : 0.0521447968456687
    fit_seconds : 24.34
    shap_seconds : 1718.39
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                  therefore understate their feature set

[record 13 of 15]
    iteration : 12
    n_features_remaining : 38

```

**Table T4. Progressive removal on PhiUSIIL: held-out performance after greedily removing the highest-attributed feature at each iteration. (continued)**

```

    removed_feature : NoOfLettersInURL
    accuracy : 0.9925671650259582
    f1 : 0.991250918319642
    mcc : 0.9848054869947998
    top_feature_after : HasCopyrightInfo
    removed_so_far : 12
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
    HasSocialNet; NoOfJS; NoOfCSS; LargestLineLength; LetterRatioInURL;
    NoOfOtherSpecialCharsInURL
    top_feature_before : NoOfLettersInURL
    top_shap_before : 0.0557818683421087
    second_feature : HasCopyrightInfo
    second_shap : 0.0509389295780166
    fit_seconds : 31.67
    shap_seconds : 1597.42
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
    therefore understate their feature set

[record 14 of 15]
    iteration : 13
    n_features_remaining : 37
    removed_feature : HasCopyrightInfo
    accuracy : 0.991801185849244
    f1 : 0.990350906480585
    mcc : 0.9832372088397836
    top_feature_after : HasDescription
    removed_so_far : 13
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
    HasSocialNet; NoOfJS; NoOfCSS; LargestLineLength; LetterRatioInURL;
    NoOfOtherSpecialCharsInURL; NoOfLettersInURL
    top_feature_before : HasCopyrightInfo
    top_shap_before : 0.0573962341788812
    second_feature : HasDescription
    second_shap : 0.0535061880736899
    fit_seconds : 24.72
    shap_seconds : 1571.49
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
    therefore understate their feature set

[record 15 of 15]
    iteration : 14
    n_features_remaining : 36
    removed_feature :
    accuracy : 0.989191182728588
    f1 : 0.9873055009495884
    mcc : 0.9778960075454448
    top_feature_after :
    removed_so_far : 14
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
    HasSocialNet; NoOfJS; NoOfCSS; LargestLineLength; LetterRatioInURL;
    NoOfOtherSpecialCharsInURL; NoOfLettersInURL; HasCopyrightInfo
    top_feature_before : HasDescription
    top_shap_before : 0.0594322166570935
    second_feature : DomainTitleMatchScore
    second_shap : 0.0528102098854716
    fit_seconds : 33.0
    shap_seconds : 2022.07
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
    therefore understate their feature set

```

**Table T4. Progressive removal on PhiUSIL: held-out performance after greedily removing the highest-attributed feature at each iteration. (continued)**

**Table T6. Zero-shot transfer between benchmarks on the shared nine-concept schema, five seeds per direction.**

```
[record 1 of 4]
    direction : phiusiil->phiusiil
      train   : phiusiil
      test    : phiusiil
in_distribution : True
  n_seeds    : 5
  n_test     : 35249
majority_class_accuracy : 0.5738602513546484
  accuracy_mean : 0.9019262957814408
  accuracy_std  : 0.0006598600442346
  accuracy_ci_lower : 0.9013929473176544
  accuracy_ci_upper : 0.9024369485659168
  precision_mean : 0.9056444339863938
  precision_std  : 0.0016384277650987
  precision_ci_lower : 0.9043577558234336
  precision_ci_upper : 0.9068350358020898
  recall_mean : 0.859396844417815
  recall_std  : 0.0017820063663303
  recall_ci_lower : 0.8580254310631782
  recall_ci_upper : 0.8607695892417283
  f1_mean : 0.8819126663106234
  f1_std  : 0.0008118634767744
  f1_ci_lower : 0.8813078511702008
  f1_ci_upper : 0.8825885014345717
  mcc_mean : 0.7989546295294948
  mcc_std  : 0.0013595192955533
  mcc_ci_lower : 0.7978481610074346
  mcc_ci_upper : 0.80001235371225
  roc_auc_mean : 0.9669932143117757
  roc_auc_std  : 0.0003512505464175
  roc_auc_ci_lower : 0.966729356897838
  roc_auc_ci_upper : 0.9672570717257128
accuracy_above_majority : 0.3280660444267923
```

```
[record 2 of 4]
    direction : phiusiil->uci
      train   : phiusiil
      test    : uci
in_distribution : False
  n_seeds    : 5
  n_test     : 1659
majority_class_accuracy : 0.5569620253164558
  accuracy_mean : 0.5265822784810126
  accuracy_std  : 0.0051782354659227
  accuracy_ci_lower : 0.5226039783001808
  accuracy_ci_upper : 0.531045810729355
  precision_mean : 0.4831216822174379
  precision_std  : 0.0027620802310054
  precision_ci_lower : 0.4810021693342975
  precision_ci_upper : 0.485517997823656
  recall_mean : 0.9804081632653062
  recall_std  : 0.003547865254532
  recall_ci_lower : 0.9779523809523808
  recall_ci_upper : 0.9834013605442176
  f1_mean : 0.6472720805963317
  f1_std  : 0.0022272565970481
  f1_ci_lower : 0.6455072676925744
  f1_ci_upper : 0.6490009463178119
  mcc_mean : 0.240783149229503
  mcc_std  : 0.0086634870694218
  mcc_ci_lower : 0.2337392084360741
  mcc_ci_upper : 0.2473856120508793
```

**Table T6. Zero-shot transfer between benchmarks on the shared nine-concept schema, five seeds per direction. (continued)**

```

roc_auc_mean : 0.8401956886650763
roc_auc_std : 0.0256361857735798
roc_auc_ci_lower : 0.818209058515181
roc_auc_ci_upper : 0.8555607091321376
accuracy_above_majority : -0.0303797468354432

```

[record 3 of 4]

```

direction : uci->uci
train : uci
test : uci
in_distribution : True
n_seeds : 5
n_test : 1659
majority_class_accuracy : 0.5569620253164558
accuracy_mean : 0.9205545509342976
accuracy_std : 0.0055146295414021
accuracy_ci_lower : 0.916576250753466
accuracy_ci_upper : 0.9250150693188668
precision_mean : 0.9284360862153792
precision_std : 0.0063231068912138
precision_ci_lower : 0.9232132415948936
precision_ci_upper : 0.933204161703942
recall_mean : 0.8892517006802722
recall_std : 0.0105650441677916
recall_ci_lower : 0.8802721088435372
recall_ci_upper : 0.8965986394557823
f1_mean : 0.9083910184922048
f1_std : 0.0066509988848052
f1_ci_lower : 0.9034051502405068
f1_ci_upper : 0.913758538162139
mcc_mean : 0.838964368316882
mcc_std : 0.0111231979990243
mcc_ci_lower : 0.8309454686098032
mcc_ci_upper : 0.8479609433037714
roc_auc_mean : 0.9740947374620844
roc_auc_std : 0.0022770168938744
roc_auc_ci_lower : 0.9724295285213652
roc_auc_ci_upper : 0.975889312218394
accuracy_above_majority : 0.3635925256178419

```

[record 4 of 4]

```

direction : uci->phiusiil
train : uci
test : phiusiil
in_distribution : False
n_seeds : 5
n_test : 35249
majority_class_accuracy : 0.5738602513546484
accuracy_mean : 0.6402053958977559
accuracy_std : 0.0113818428681579
accuracy_ci_lower : 0.6300434054866805
accuracy_ci_upper : 0.6468155124968085
precision_mean : 0.859368831430721
precision_std : 0.0165619140936406
precision_ci_lower : 0.8468175896656912
precision_ci_upper : 0.8729784745467195
recall_mean : 0.1859130550562545
recall_std : 0.0298115941240398
recall_ci_lower : 0.1598828307036815
recall_ci_upper : 0.2037014845882431
f1_mean : 0.3048298427050162
f1_std : 0.0420244690085735

```

**Table T6. Zero-shot transfer between benchmarks on the shared nine-concept schema, five seeds per direction. (continued)**

```
f1_ci_lower : 0.2678617797510827
f1_ci_upper : 0.3293774311645302
  mcc_mean : 0.2787652457513706
  mcc_std : 0.0297141582331254
mcc_ci_lower : 0.2521239485140789
mcc_ci_upper : 0.2961691812424171
  roc_auc_mean : 0.7144176424049767
  roc_auc_std : 0.0161339241007508
roc_auc_ci_lower : 0.7040834358494903
roc_auc_ci_upper : 0.7283761239883436
accuracy_above_majority : 0.0663451445431075
```

**Table T7. Within-dataset ceiling: Kendall's tau between attribution rankings from independent models on disjoint halves of the same data.**

[record 1 of 4]

```
config : phiusiil_full
dataset : phiusiil
n_seeds : 5
n_features : 50
top1_agreement_rate : 1.0
n_features_sign_top10_mean : 9.2
kendall_tau_mean : 0.9190204081632654
kendall_tau_std : 0.0175310755207132
kendall_tau_ci_lower : 0.9053061224489796
kendall_tau_ci_upper : 0.9327346938775511
jaccard_5_mean : 0.9333333333333332
jaccard_5_std : 0.149071198499986
jaccard_5_ci_lower : 0.7999999999999999
jaccard_5_ci_upper : 1.0
jaccard_10_mean : 0.8606060606060606
jaccard_10_std : 0.1414862730900455
jaccard_10_ci_lower : 0.7575757575757576
jaccard_10_ci_upper : 0.9636363636363636
jaccard_15_mean : 0.8529411764705882
jaccard_15_std : 0.0493250289154365
jaccard_15_ci_lower : 0.8088235294117647
jaccard_15_ci_upper : 0.875
sign_agreement_top5_mean : 34.0
sign_agreement_top5_std : 24.08318915758459
sign_agreement_top5_ci_lower : 12.0
sign_agreement_top5_ci_upper : 52.0
sign_agreement_top10_mean : 41.22222222222221
sign_agreement_top10_std : 19.02889291136241
sign_agreement_top10_ci_lower : 26.866666666666667
sign_agreement_top10_ci_upper : 55.58333333333331
sign_agreement_top15_mean : 41.86813186813187
sign_agreement_top15_std : 13.251899814357632
sign_agreement_top15_ci_lower : 33.73626373626374
sign_agreement_top15_ci_upper : 54.28571428571429
sign_agreement_weighted_mean : 26.24985992335248
sign_agreement_weighted_std : 30.18856396281425
sign_agreement_weighted_ci_lower : 5.249993965166468
sign_agreement_weighted_ci_upper : 52.81873711378464
sign_agreement_unfiltered_mean : 43.6
sign_agreement_unfiltered_std : 11.260550608207398
sign_agreement_unfiltered_ci_lower : 36.0
sign_agreement_unfiltered_ci_upper : 53.2
```

[record 2 of 4]

```
config : phiusiil_leakfree
dataset : phiusiil
n_seeds : 5
n_features : 36
top1_agreement_rate : 0.8
n_features_sign_top10_mean : 9.6
kendall_tau_mean : 0.9307936507936508
kendall_tau_std : 0.0098872453983475
kendall_tau_ci_lower : 0.9225396825396828
kendall_tau_ci_upper : 0.9371428571428572
jaccard_5_mean : 0.8666666666666666
jaccard_5_std : 0.1825741858350553
jaccard_5_ci_lower : 0.7333333333333332
jaccard_5_ci_upper : 1.0
jaccard_10_mean : 0.9272727272727274
jaccard_10_std : 0.0995859195463938
jaccard_10_ci_lower : 0.8545454545454545
```

**Table T7. Within-dataset ceiling: Kendall's tau between attribution rankings from independent models on disjoint halves of the same data. (continued)**

```

jaccard_10_ci_upper : 1.0
jaccard_15_mean : 0.9
jaccard_15_std : 0.0559016994374947
jaccard_15_ci_lower : 0.875
jaccard_15_ci_upper : 0.95
sign_agreement_top5_mean : 62.0
sign_agreement_top5_std : 30.331501776206203
sign_agreement_top5_ci_lower : 38.0
sign_agreement_top5_ci_upper : 84.04999999999995
sign_agreement_top10_mean : 51.333333333333336
sign_agreement_top10_std : 21.17534088812728
sign_agreement_top10_ci_lower : 35.33333333333333
sign_agreement_top10_ci_upper : 68.0
sign_agreement_top15_mean : 44.0
sign_agreement_top15_std : 16.79407086821211
sign_agreement_top15_ci_lower : 32.0
sign_agreement_top15_ci_upper : 55.71428571428571
sign_agreement_weighted_mean : 47.13274094309819
sign_agreement_weighted_std : 19.149237951466915
sign_agreement_weighted_ci_lower : 33.085354004943284
sign_agreement_weighted_ci_upper : 61.737679147962425
sign_agreement_unfiltered_mean : 48.88888888888889
sign_agreement_unfiltered_std : 11.719457283182768
sign_agreement_unfiltered_ci_lower : 40.55555555555556
sign_agreement_unfiltered_ci_upper : 58.88888888888888

```

[record 3 of 4]

```

config : uci_dedup
dataset : uci
n_seeds : 5
n_features : 30
top1_agreement_rate : 0.4
n_features_sign_top10_mean : 9.0
kendall_tau_mean : 0.7682758620689655
kendall_tau_std : 0.0115401382970217
kendall_tau_ci_lower : 0.759080459770115
kendall_tau_ci_upper : 0.7765747126436782
jaccard_5_mean : 0.6666666666666666
jaccard_5_std : 0.0
jaccard_5_ci_lower : 0.6666666666666666
jaccard_5_ci_upper : 0.6666666666666666
jaccard_10_mean : 0.8181818181818181
jaccard_10_std : 1.2412670766236366e-16
jaccard_10_ci_lower : 0.8181818181818181
jaccard_10_ci_upper : 0.8181818181818181
jaccard_15_mean : 0.7892156862745098
jaccard_15_std : 0.0879454662647507
jaccard_15_ci_lower : 0.7254901960784312
jaccard_15_ci_upper : 0.8529411764705882
sign_agreement_top5_mean : 50.0
sign_agreement_top5_std : 17.67766952966369
sign_agreement_top5_ci_lower : 35.0
sign_agreement_top5_ci_upper : 65.0
sign_agreement_top10_mean : 44.44444444444444
sign_agreement_top10_std : 17.568209223157663
sign_agreement_top10_ci_lower : 31.111111111111107
sign_agreement_top10_ci_upper : 57.77777777777777
sign_agreement_top15_mean : 51.684981684981686
sign_agreement_top15_std : 12.909424815646569
sign_agreement_top15_ci_lower : 41.75824175824176
sign_agreement_top15_ci_upper : 61.61172161172161
sign_agreement_weighted_mean : 55.69565427371174

```

**Table T7. Within-dataset ceiling: Kendall's tau between attribution rankings from independent models on disjoint halves of the same data. (continued)**

```
sign_agreement_weighted_std : 13.867915059366505
sign_agreement_weighted_ci_lower : 45.54650394227103
sign_agreement_weighted_ci_upper : 67.4256585842625
sign_agreement_unfiltered_mean : 50.66666666666667
sign_agreement_unfiltered_std : 4.944132324730441
sign_agreement_unfiltered_ci_lower : 46.66666666666667
sign_agreement_unfiltered_ci_upper : 54.0
```

[record 4 of 4]

```
config : uci_full
dataset : uci
n_seeds : 5
n_features : 30
top1_agreement_rate : 0.6
n_features_sign_top10_mean : 9.6
kendall_tau_mean : 0.824367816091954
kendall_tau_std : 0.0185053901586189
kendall_tau_ci_lower : 0.8096321839080459
kendall_tau_ci_upper : 0.83816091954023
jaccard_5_mean : 0.8
jaccard_5_std : 0.1825741858350553
jaccard_5_ci_lower : 0.6666666666666666
jaccard_5_ci_upper : 0.9333333333333332
jaccard_10_mean : 0.9272727272727274
jaccard_10_std : 0.0995859195463938
jaccard_10_ci_lower : 0.8545454545454547
jaccard_10_ci_upper : 1.0
jaccard_15_mean : 0.9
jaccard_15_std : 0.0559016994374947
jaccard_15_ci_lower : 0.875
jaccard_15_ci_upper : 0.95
sign_agreement_top5_mean : 45.0
sign_agreement_top5_std : 13.228756555322953
sign_agreement_top5_ci_lower : 33.0
sign_agreement_top5_ci_upper : 54.0
sign_agreement_top10_mean : 44.0
sign_agreement_top10_std : 15.666272650679616
sign_agreement_top10_ci_lower : 32.0
sign_agreement_top10_ci_upper : 56.0
sign_agreement_top15_mean : 46.38095238095239
sign_agreement_top15_std : 11.256643289213246
sign_agreement_top15_ci_lower : 37.14285714285714
sign_agreement_top15_ci_upper : 54.095238095238095
sign_agreement_weighted_mean : 47.14090504304773
sign_agreement_weighted_std : 24.162626591280908
sign_agreement_weighted_ci_lower : 26.45745296751132
sign_agreement_weighted_ci_upper : 64.92344984195503
sign_agreement_unfiltered_mean : 46.0
sign_agreement_unfiltered_std : 4.346134936801765
sign_agreement_unfiltered_ci_lower : 42.666666666666666
sign_agreement_unfiltered_ci_upper : 49.33333333333333
```

**Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement.**

```
[record 1 of 69]
  concept : ip_in_domain
  status : mapped
  uci_feature : having_ip_address
  phiusiil_source : IsDomainIP
  rationale : Both encode 'the host is a bare IP address rather than a domain name'. Direct binary correspondence.
  caveat :
  rho_phiusiil : -0.0605489999124573
  rho_uci : -0.0941600949562038
direction_agrees : True
polarity_status : agrees

[record 2 of 69]
  concept : url_length
  status : mapped
  uci_feature : url_length
  phiusiil_source : URLLength
  rationale : Same quantity (characters in the URL); PhiUSIIL stores it raw, UCI stores it binned. Discretised with UCI's published 54/75 thresholds.
  caveat :
  rho_phiusiil : -0.3539937654486287
  rho_uci : -0.0533315794339325
direction_agrees : True
polarity_status : agrees

[record 3 of 69]
  concept : subdomain_depth
  status : mapped
  uci_feature : having_sub_domain
  phiusiil_source : NoOfSubDomain
  rationale : Both encode subdomain nesting depth. UCI counts dots after stripping www and ccTLD; PhiUSIIL counts subdomains directly, so the counts align up to the stripping convention.
  caveat : UCI's dot-count and PhiUSIIL's subdomain count can differ by one for hosts with a ccTLD (e.g. .co.uk).
  rho_phiusiil : -0.1141491829083289
  rho_uci : -0.3046922676521399
direction_agrees : True
polarity_status : agrees

[record 4 of 69]
  concept : ssl_state
  status : mapped
  uci_feature : sslfinal_state
  phiusiil_source : IsHTTPS
  rationale : Closest available correspondence to UCI's SSL assessment: whether the page is served over HTTPS at all, which is the dominant component of sslfinal_state.
  caveat : IMPERFECT. UCI's sslfinal_state additionally requires a trusted issuer and a certificate at least a year old, and has a 'suspicious' state for https-with-untrusted-issuer. PhiUSIIL records only the scheme, so the suspicious state is unreachable from the PhiUSIIL side.
  rho_phiusiil : -0.6124154229775562
  rho_uci : -0.735813975062039
direction_agrees : True
polarity_status : agrees

[record 5 of 69]
  concept : iframe
  status : mapped
  uci_feature : iframe
  phiusiil_source : NoOfiFrame
  rationale : Both encode iframe usage on the page.
  caveat : UCI targets specifically INVISIBLE iframes; PhiUSIIL counts all iframes. Note also that UCI's
```

**Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)**

```

        iframe feature carries no marginal signal at all (phishing rate 0.44 at both of its values), so
        this concept contributes nothing on the UCI side regardless of the mapping quality.
    rho_phiusiil : 0.521331928092635
    rho_uci : 0.0033935238644643
    direction_agrees : False
    polarity_status : NO_SIGNAL_UCI

[record 6 of 69]
    concept : popup
    status : mapped
    uci_feature : popupwindow
    phiusiil_source : NoOfPopup
    rationale : Both encode popup-window usage.
    caveat : UCI targets popups containing a text field; PhiUSIIL counts all popups. As with iframe, UCI's
        popupwindow has phishing rate 0.44 at both values - zero marginal signal on the UCI side.
    rho_phiusiil : 0.1925460026806499
    rho_uci : -8.588679256217027e-05
    direction_agrees : False
    polarity_status : NO_SIGNAL_UCI

[record 7 of 69]
    concept : url_redirect
    status : mapped
    uci_feature : redirect
    phiusiil_source : NoOfURLRedirect
    rationale : Both encode redirection on the way to the page. UCI's `redirect` takes only {0, 1} in the delivered
        data, so the PhiUSIIL side is mapped onto that same {0, 1} domain rather than the usual ternary
        one.
    caveat : UCI's published rule describes a 3-way split on redirect count, but the delivered column is binary;
        the mapping follows the DATA, not the paper.
    rho_phiusiil : 0.0480143528227923
    rho_uci : 0.0201134616575737
    direction_agrees : True
    polarity_status : agrees

[record 8 of 69]
    concept : form_handler
    status : mapped
    uci_feature : sfh
    phiusiil_source : HasExternalFormSubmit
    rationale : Both encode whether a form posts to a different domain than the page - UCI's Server Form Handler
        check and PhiUSIIL's external-form-submit flag.
    caveat : UCI's 'suspicious' state (blank/about:blank handler) has no PhiUSIIL counterpart and is unreachable
        from the PhiUSIIL side.
    rho_phiusiil : 0.1673089014798561
    rho_uci : -0.2191900653063038
    direction_agrees : False
    polarity_status : INVERTED

[record 9 of 69]
    concept : anchor_external_ratio
    status : mapped
    uci_feature : url_of_anchor
    phiusiil_source : NoOfExternalRef + NoOfSelfRef + NoOfEmptyRef
    rationale : UCI's url_of_anchor is the share of <a> hrefs pointing off-domain. PhiUSIIL provides the underlying
        counts (external / self / empty references), so the ratio is reconstructed directly and binned with
        UCI's 31%/67% thresholds.
    caveat :
    rho_phiusiil : 0.0872798521005418
    rho_uci : -0.7012072591666815
    direction_agrees : False
    polarity_status : INVERTED

```

**Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)**

```
[record 10 of 69]
  concept :
    status : unmapped_uci
    uci_feature : shortining_service
  phiusiil_source :
    rationale : URL-shortener detection; PhiUSIIL has no shortener feature.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 11 of 69]
  concept :
    status : unmapped_uci
    uci_feature : having_at_symbol
  phiusiil_source :
    rationale : presence of '@' in the URL; PhiUSIIL counts only aggregate special characters, with no per-
    character breakdown.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 12 of 69]
  concept :
    status : unmapped_uci
    uci_feature : double_slash_redirecting
  phiusiil_source :
    rationale : '//' position in the path; no PhiUSIIL equivalent.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 13 of 69]
  concept :
    status : unmapped_uci
    uci_feature : prefix_suffix
  phiusiil_source :
    rationale : hyphen in the domain; PhiUSIIL has no hyphen feature.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 14 of 69]
  concept :
    status : unmapped_uci
    uci_feature : domain_registration_length
  phiusiil_source :
    rationale : WHOIS registration length; PhiUSIIL has no WHOIS data.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :
```

**Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)**

```
[record 15 of 69]
  concept :
    status : unmapped_uci
  uci_feature : favicon
  phiusiil_source :
    rationale : favicon loaded from an EXTERNAL domain. PhiUSIIL's HasFavicon records presence only, which is a
                different concept - mapping them would force a match between 'has a favicon' and 'favicon is
                foreign'.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 16 of 69]
  concept :
    status : unmapped_uci
  uci_feature : port
  phiusiil_source :
    rationale : non-standard port; no PhiUSIIL equivalent.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 17 of 69]
  concept :
    status : unmapped_uci
  uci_feature : https_token
  phiusiil_source :
    rationale : the literal token 'https' inside the domain name (a deception pattern), NOT whether HTTPS is used.
                PhiUSIIL's IsHTTPS is the latter and is already mapped to sslfinal_state.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 18 of 69]
  concept :
    status : unmapped_uci
  uci_feature : request_url
  phiusiil_source :
    rationale : share of loaded OBJECTS (images/scripts) from an external domain. PhiUSIIL has
                NoOfImage/NoOfCSS/NoOfJS but no internal/external split for them, so the ratio cannot be
                reconstructed.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 19 of 69]
  concept :
    status : unmapped_uci
  uci_feature : links_in_tags
  phiusiil_source :
    rationale : share of links inside meta/script/link tags; PhiUSIIL has no tag-scoped link counts.
    caveat :
  rho_phiusiil :
  rho_uci :
```

**Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)**

```

direction_agrees :
polarity_status :

[record 20 of 69]
  concept :
    status : unmapped_uci
  uci_feature : submitting_to_email
  phiusiil_source :
    rationale : mailto: form action; no PhiUSIIL equivalent.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 21 of 69]
  concept :
    status : unmapped_uci
  uci_feature : abnormal_url
  phiusiil_source :
    rationale : WHOIS hostname consistency; no PhiUSIIL WHOIS data.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 22 of 69]
  concept :
    status : unmapped_uci
  uci_feature : on_mouseover
  phiusiil_source :
    rationale : JavaScript status-bar manipulation; PhiUSIIL records no JS behaviour.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 23 of 69]
  concept :
    status : unmapped_uci
  uci_feature : rightclick
  phiusiil_source :
    rationale : right-click disabling; PhiUSIIL records no JS behaviour.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 24 of 69]
  concept :
    status : unmapped_uci
  uci_feature : age_of_domain
  phiusiil_source :
    rationale : WHOIS domain age; no PhiUSIIL WHOIS data.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

```

**Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)**

```
[record 25 of 69]
  concept :
    status : unmapped_uci
    uci_feature : dnsrecord
  phiusiil_source :
    rationale : DNS record existence; no PhiUSIIL DNS data.
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 26 of 69]
  concept :
    status : unmapped_uci
    uci_feature : web_traffic
  phiusiil_source :
    rationale : Alexa traffic rank; external reputation, absent from PhiUSIIL.
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 27 of 69]
  concept :
    status : unmapped_uci
    uci_feature : page_rank
  phiusiil_source :
    rationale : PageRank; external reputation, absent from PhiUSIIL.
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 28 of 69]
  concept :
    status : unmapped_uci
    uci_feature : google_index
  phiusiil_source :
    rationale : Google index membership; external reputation, absent from PhiUSIIL.
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 29 of 69]
  concept :
    status : unmapped_uci
    uci_feature : links_pointing_to_page
  phiusiil_source :
    rationale : INBOUND link count. PhiUSIIL's NoOfExternalRef is OUTBOUND; mapping them would invert the direction
    of the relationship.
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :
```

**Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)**

```
[record 30 of 69]
  concept :
    status : unmapped_uci
  uci_feature : statistical_report
  phiusiil_source :
    rationale : membership of a published blacklist; external reputation.
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 31 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : DomainLength
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 32 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : URLSimilarityIndex
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 33 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : CharContinuationRate
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 34 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : TLDLegitimateProb
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 35 of 69]
  concept :
```

**Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)**

```

    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : URLCharProb
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 36 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : TLDLength
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 37 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : HasObfuscation
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 38 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : NoOf0bfuscatedChar
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 39 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : ObfuscationRatio
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 40 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :

```

**Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)**

```

    phiusiil_source : NoOfLettersInURL
      rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
      caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 41 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
    phiusiil_source : LetterRatioInURL
      rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
      caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 42 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
    phiusiil_source : NoOfDegitsInURL
      rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
      caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 43 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
    phiusiil_source : DgitRatioInURL
      rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
      caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 44 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
    phiusiil_source : NoOfEqualsInURL
      rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
      caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 45 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
    phiusiil_source : NoOfQMarkInURL
      rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)

```

**Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)**

```

    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 46 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : NoOfAmpersandInURL
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 47 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : NoOfOtherSpecialCharsInURL
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 48 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : SpacialCharRatioInURL
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 49 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : LineOfCode
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 50 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : LargestLineLength
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :

```

**Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)**

|                    |                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------|
| rho_uci :          |                                                                                            |
| direction_agrees : |                                                                                            |
| polarity_status :  |                                                                                            |
| [record 51 of 69]  |                                                                                            |
| concept :          |                                                                                            |
| status :           | unmapped_phiusiil                                                                          |
| uci_feature :      |                                                                                            |
| phiusiil_source :  | HasTitle                                                                                   |
| rationale :        | no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature) |
| caveat :           |                                                                                            |
| rho_phiusiil :     |                                                                                            |
| rho_uci :          |                                                                                            |
| direction_agrees : |                                                                                            |
| polarity_status :  |                                                                                            |
| [record 52 of 69]  |                                                                                            |
| concept :          |                                                                                            |
| status :           | unmapped_phiusiil                                                                          |
| uci_feature :      |                                                                                            |
| phiusiil_source :  | DomainTitleMatchScore                                                                      |
| rationale :        | no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature) |
| caveat :           |                                                                                            |
| rho_phiusiil :     |                                                                                            |
| rho_uci :          |                                                                                            |
| direction_agrees : |                                                                                            |
| polarity_status :  |                                                                                            |
| [record 53 of 69]  |                                                                                            |
| concept :          |                                                                                            |
| status :           | unmapped_phiusiil                                                                          |
| uci_feature :      |                                                                                            |
| phiusiil_source :  | URLTitleMatchScore                                                                         |
| rationale :        | no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature) |
| caveat :           |                                                                                            |
| rho_phiusiil :     |                                                                                            |
| rho_uci :          |                                                                                            |
| direction_agrees : |                                                                                            |
| polarity_status :  |                                                                                            |
| [record 54 of 69]  |                                                                                            |
| concept :          |                                                                                            |
| status :           | unmapped_phiusiil                                                                          |
| uci_feature :      |                                                                                            |
| phiusiil_source :  | HasFavicon                                                                                 |
| rationale :        | no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature) |
| caveat :           |                                                                                            |
| rho_phiusiil :     |                                                                                            |
| rho_uci :          |                                                                                            |
| direction_agrees : |                                                                                            |
| polarity_status :  |                                                                                            |
| [record 55 of 69]  |                                                                                            |
| concept :          |                                                                                            |
| status :           | unmapped_phiusiil                                                                          |
| uci_feature :      |                                                                                            |
| phiusiil_source :  | Robots                                                                                     |
| rationale :        | no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature) |
| caveat :           |                                                                                            |
| rho_phiusiil :     |                                                                                            |
| rho_uci :          |                                                                                            |
| direction_agrees : |                                                                                            |

**Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)**

```

polarity_status :

[record 56 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : IsResponsive
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 57 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : NoOfSelfRedirect
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 58 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : HasDescription
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 59 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : HasSocialNet
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 60 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : HasSubmitButton
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

```

**Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)**

```
[record 61 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : HasHiddenFields
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 62 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : HasPasswordField
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 63 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : Bank
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 64 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : Pay
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 65 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : Crypto
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 66 of 69]
  concept :
```

**Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)**

```

    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : HasCopyrightInfo
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 67 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : NoOfImage
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 68 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : NoOfCSS
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 69 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : NoOfJS
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

```

**Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load.**

[record 1 of 24]

```

        dataset : phiusiil
        iteration : 0
        n_features : 50
        accuracy : 1.0
        fit_seconds : 20.46
        shap_seconds : 284.19
        mean_tree_depth : 13.864
        total_nodes : 104862
        window_start : 10:34:34
        window_end : 10:38:27
        cache_replay : False
        source_log : a13.log
        run_date : 2026-09-11
        window_reconstructed : True
        competing_jobs :
        contention_evidence : remeasured_uncontended
        sampled_cpu_percent_mean :
        sampled_cpu_percent_max :
        sampled_n_python_procs_max :
        sampled_job_tags :
        n_load_samples :
        contended : False
        cost_analysis_eligible : True
        remeasured_clean : True
        shap_seconds_original_contended : 179.9
        f1 :
        mcc :
        removed_feature :
        top_feature :
        hyperparameter_regime :

```

[record 2 of 24]

```

        dataset : phiusiil
        iteration : 1
        n_features : 49
        accuracy : 0.999716
        fit_seconds : 31.2
        shap_seconds : 683.21
        mean_tree_depth : 19.158
        total_nodes : 281270
        window_start : 10:38:27
        window_end : 10:46:39
        cache_replay : False
        source_log : a13.log
        run_date : 2026-09-11
        window_reconstructed :
        competing_jobs :
        contention_evidence : remeasured_uncontended
        sampled_cpu_percent_mean :
        sampled_cpu_percent_max :
        sampled_n_python_procs_max :
        sampled_job_tags :
        n_load_samples :
        contended : False
        cost_analysis_eligible : True
        remeasured_clean : True
        shap_seconds_original_contended : 463.5
        f1 :
        mcc :
        removed_feature :
        top_feature :

```

**Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)**

```

hyperparameter_regime :
[record 3 of 24]
    dataset : phiusiil
    iteration : 2
    n_features : 48
    accuracy : 0.999546
    fit_seconds : 27.4
    shap_seconds : 788.7
    mean_tree_depth : 20.152
    total_nodes : 367168
    window_start : 10:46:39
    window_end : 10:56:32
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed :
    competing_jobs :
    contention_evidence : remeasured_uncontended
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
    contended : False
    cost_analysis_eligible : True
    remeasured_clean : True
shap_seconds_original_contended : 564.3
    f1 :
    mcc :
    removed_feature :
    top_feature :
hyperparameter_regime :

```

```

[record 4 of 24]
    dataset : phiusiil
    iteration : 3
    n_features : 47
    accuracy : 0.998156
    fit_seconds : 28.77
    shap_seconds : 978.95
    mean_tree_depth : 23.722
    total_nodes : 544400
    window_start : 10:56:32
    window_end : 11:08:45
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed :
    competing_jobs :
    contention_evidence : remeasured_uncontended
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
    contended : False
    cost_analysis_eligible : True
    remeasured_clean : True
shap_seconds_original_contended : 705.8
    f1 :
    mcc :

```

**Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)**

```

        removed_feature :
        top_feature :
    hyperparameter_regime :

[record 5 of 24]
        dataset : phiusiil
        iteration : 4
        n_features : 46
        accuracy : 0.997844
        fit_seconds : 27.9
        shap_seconds : 582.7
    mean_tree_depth : 25.126
        total_nodes : 628132
        window_start : 11:08:45
        window_end : 11:18:56
        cache_replay : False
        source_log : a13.log
        run_date : 2026-09-11
    window_reconstructed :
        competing_jobs :
        contention_evidence :
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
        sampled_job_tags :
        n_load_samples :
            contended : False
    cost_analysis_eligible : True
        remeasured_clean : False
    shap_seconds_original_contended :
        f1 :
        mcc :
        removed_feature :
        top_feature :
    hyperparameter_regime :

[record 6 of 24]
        dataset : phiusiil
        iteration : 5
        n_features : 45
        accuracy : 0.99773
        fit_seconds : 25.2
        shap_seconds : 851.9
    mean_tree_depth : 26.246
        total_nodes : 704950
        window_start : 11:18:56
        window_end : 11:33:34
        cache_replay : False
        source_log : a13.log
        run_date : 2026-09-11
    window_reconstructed :
        competing_jobs :
        contention_evidence :
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
        sampled_job_tags :
        n_load_samples :
            contended : False
    cost_analysis_eligible : True
        remeasured_clean : False
    shap_seconds_original_contended :

```

**Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)**

```

        f1 :
        mcc :
    removed_feature :
        top_feature :
    hyperparameter_regime :

[record 7 of 24]
    dataset : phiusiil
    iteration : 6
    n_features : 44
    accuracy : 0.997362
    fit_seconds : 25.4
    shap_seconds : 752.1
    mean_tree_depth : 28.38
    total_nodes : 838026
    window_start : 11:33:34
    window_end : 11:46:32
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed :
    competing_jobs :
    contention_evidence :
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
    contended : False
    cost_analysis_eligible : True
    remeasured_clean : False
shap_seconds_original_contended :
    f1 :
    mcc :
    removed_feature :
    top_feature :
    hyperparameter_regime :

```

```

[record 8 of 24]
    dataset : phiusiil
    iteration : 7
    n_features : 43
    accuracy : 0.996993
    fit_seconds : 32.3
    shap_seconds : 700.5
    mean_tree_depth : 28.88
    total_nodes : 952992
    window_start : 11:46:32
    window_end : 11:58:46
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed :
    competing_jobs :
    contention_evidence :
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
    contended : False
    cost_analysis_eligible : True

```

**Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)**

```

remeasured_clean : False
shap_seconds_original_contended :
    f1 :
    mcc :
    removed_feature :
    top_feature :
hyperparameter_regime :

[record 9 of 24]
    dataset : phiusiil
    iteration : 8
    n_features : 42
    accuracy : 0.996567
    fit_seconds : 24.1
    shap_seconds : 717.8
    mean_tree_depth : 31.152
    total_nodes : 1075554
    window_start : 11:58:46
    window_end : 12:11:08
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
window_reconstructed :
    competing_jobs :
    contention_evidence :
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
    contended : False
    cost_analysis_eligible : True
    remeasured_clean : False
shap_seconds_original_contended :
    f1 :
    mcc :
    removed_feature :
    top_feature :
hyperparameter_regime :
```

```

[record 10 of 24]
    dataset : phiusiil
    iteration : 9
    n_features : 41
    accuracy : 0.995376
    fit_seconds : 29.1
    shap_seconds : 1220.1
    mean_tree_depth : 33.43
    total_nodes : 1291626
    window_start : 12:11:08
    window_end : 12:31:58
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
window_reconstructed :
    competing_jobs :
    contention_evidence :
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
```

**Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)**

```

        contended : False
cost_analysis_eligible : True
    remeasured_clean : False
shap_seconds_original_contended :
    f1 :
    mcc :
    removed_feature :
    top_feature :
hyperparameter_regime :

```

[record 11 of 24]

```

    dataset : phiusiil
    iteration : 10
    n_features : 40
    accuracy : 0.994581
    fit_seconds : 25.4
    shap_seconds : 1264.2
    mean_tree_depth : 36.51
    total_nodes : 1496914
    window_start : 12:31:58
    window_end : 12:53:28
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
window_reconstructed :
    competing_jobs :
    contention_evidence :
sampled_cpu_percent_mean :
sampled_cpu_percent_max :
sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
        contended : False
cost_analysis_eligible : True
    remeasured_clean : False
shap_seconds_original_contended :
    f1 :
    mcc :
    removed_feature :
    top_feature :
hyperparameter_regime :

```

[record 12 of 24]

```

    dataset : phiusiil
    iteration : 11
    n_features : 39
    accuracy : 0.992766
    fit_seconds : 28.83
    shap_seconds : 1860.9
    mean_tree_depth : 39.466
    total_nodes : 1718032
    window_start : 14:59:46
    window_end : 15:28:50
    cache_replay : False
    source_log : a13_resume.log
    run_date : 2026-09-11
window_reconstructed :
    competing_jobs :
    contention_evidence : remeasured_uncontended
sampled_cpu_percent_mean :
sampled_cpu_percent_max :
sampled_n_python_procs_max :

```

**Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)**

```

sampled_job_tags :
  n_load_samples :
    contended : False
cost_analysis_eligible : True
remeasured_clean : True
shap_seconds_original_contended : 1718.4
  f1 :
  mcc :
  removed_feature :
  top_feature :
hyperparameter_regime :

[record 13 of 24]
  dataset : phiusiil
  iteration : 12
  n_features : 38
  accuracy : 0.992567
  fit_seconds : 31.7
  shap_seconds : 1597.4
  mean_tree_depth : 39.31
  total_nodes : 1738726
  window_start : 15:28:50
  window_end : 15:56:00
  cache_replay : False
  source_log : a13_resume.log
  run_date : 2026-09-11
  window_reconstructed :
  competing_jobs : B1;B2;B2acq
  contention_evidence : phase_a.log+sampler
  sampled_cpu_percent_mean : 38.08
  sampled_cpu_percent_max : 100.0
  sampled_n_python_procs_max : 53.0
  sampled_job_tags : B2
  n_load_samples : 46.0
  contended : True
  cost_analysis_eligible : False
  remeasured_clean : False
shap_seconds_original_contended :
  f1 :
  mcc :
  removed_feature :
  top_feature :
hyperparameter_regime :

[record 14 of 24]
  dataset : phiusiil
  iteration : 13
  n_features : 37
  accuracy : 0.991801
  fit_seconds : 24.7
  shap_seconds : 1571.5
  mean_tree_depth : 39.632
  total_nodes : 1866634
  window_start : 15:56:00
  window_end : 16:22:37
  cache_replay : False
  source_log : a13_resume.log
  run_date : 2026-09-11
  window_reconstructed :
  competing_jobs : B2;B2acq
  contention_evidence : phase_a.log+sampler
  sampled_cpu_percent_mean : 19.24

```

**Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)**

```

    sampled_cpu_percent_max : 100.0
    sampled_n_python_procs_max : 49.0
        sampled_job_tags : B2
            n_load_samples : 76.0
                contended : True
    cost_analysis_eligible : False
        remeasured_clean : False
shap_seconds_original_contended :
    f1 :
        mcc :
            removed_feature :
                top_feature :
                    hyperparameter_regime :

[record 15 of 24]
    dataset : phiusiil
    iteration : 14
    n_features : 36
    accuracy : 0.989191
    fit_seconds : 33.0
    shap_seconds : 2022.1
    mean_tree_depth : 42.442
    total_nodes : 2247592
    window_start : 16:22:37
    window_end : 16:56:53
    cache_replay : False
    source_log : a13_resume.log
    run_date : 2026-09-11
    window_reconstructed :
        competing_jobs :
            contention_evidence :
                sampled_cpu_percent_mean : 12.92
                sampled_cpu_percent_max : 100.0
            sampled_n_python_procs_max : 25.0
                sampled_job_tags :
                    n_load_samples : 97.0
                        contended : False
    cost_analysis_eligible : True
        remeasured_clean : False
shap_seconds_original_contended :
    f1 :
        mcc :
            removed_feature :
                top_feature :
                    hyperparameter_regime :

[record 16 of 24]
    dataset : uci379_website_phishing
    iteration : 0
    n_features : 9
    accuracy : 0.9064039408866996
    fit_seconds : 1.77
    shap_seconds : 0.01
    mean_tree_depth : 11.693333333333332
    total_nodes : 54348
    window_start :
    window_end :
    cache_replay : False
    source_log : screening
    run_date :
    window_reconstructed :
        competing_jobs :
```

**Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)**

```

    contention_evidence : not_assessed
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
    contended :
    cost_analysis_eligible :
    remeasured_clean :
shap_seconds_original_contended :
    f1 : 0.9243027888446216
    mcc : 0.8056806781916287
    removed_feature :
    top_feature : SFH
    hyperparameter_regime : retuned_per_feature_set

[record 17 of 24]
    dataset : uci379_website_phishing__suspicious_dropped
    iteration : 0
    n_features : 9
    accuracy : 0.9202127659574468
    fit_seconds : 1.23
    shap_seconds : 0.01
    mean_tree_depth : 9.48
    total_nodes : 9690
    window_start :
    window_end :
    cache_replay : False
    source_log : screening
    run_date :
    window_reconstructed :
    competing_jobs :
    contention_evidence : not_assessed
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
    contended :
    cost_analysis_eligible :
    remeasured_clean :
shap_seconds_original_contended :
    f1 : 0.9302325581395348
    mcc : 0.8375517998895313
    removed_feature :
    top_feature : SFH
    hyperparameter_regime : retuned_per_feature_set

[record 18 of 24]
    dataset : mendeley_hannousse
    iteration : 0
    n_features : 87
    accuracy : 0.965597667638484
    fit_seconds : 2.96
    shap_seconds : 0.01
    mean_tree_depth : 21.58
    total_nodes : 472714
    window_start :
    window_end :
    cache_replay : False
    source_log : screening
    run_date :

```

**Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)**

```

window_reconstructed :
  competing_jobs :
    contention_evidence : not_assessed
sampled_cpu_percent_mean :
sampled_cpu_percent_max :
sampled_n_python_procs_max :
  sampled_job_tags :
    n_load_samples :
      contended :
cost_analysis_eligible :
  remeasured_clean :
shap_seconds_original_contended :
  f1 : 0.9658762290341236
  mcc : 0.9313385142064948
  removed_feature :
    top_feature : page_rank
hyperparameter_regime : retuned_per_feature_set

```

[record 19 of 24]

```

dataset : mendeley_tan
iteration : 0
n_features : 48
  accuracy : 0.9873333333333332
fit_seconds : 1.9
shap_seconds : 0.01
mean_tree_depth : 19.966666666666665
total_nodes : 234594
window_start :
window_end :
cache_replay : False
source_log : screening
run_date :
window_reconstructed :
  competing_jobs :
    contention_evidence : not_assessed
sampled_cpu_percent_mean :
sampled_cpu_percent_max :
sampled_n_python_procs_max :
  sampled_job_tags :
    n_load_samples :
      contended :
cost_analysis_eligible :
  remeasured_clean :
shap_seconds_original_contended :
  f1 : 0.9873248832555036
  mcc : 0.9746675330381922
  removed_feature : PctExtHyperlinks
  top_feature : PctExtHyperlinks
hyperparameter_regime : retuned_per_feature_set

```

[record 20 of 24]

```

dataset : mendeley_tan
iteration : 1
n_features : 47
  accuracy : 0.9826666666666668
fit_seconds : 1.85
shap_seconds : 338.12
mean_tree_depth : 20.0
total_nodes : 271312
window_start :
window_end :
cache_replay : False

```

**Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)**

```

        source_log : screening
        run_date :
    window_reconstructed :
        competing_jobs :
        contention_evidence : not_assessed
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
        sampled_job_tags :
        n_load_samples :
        contended :
    cost_analysis_eligible :
        remeasured_clean :
    shap_seconds_original_contended :
        f1 : 0.9826666666666668
        mcc : 0.9653333333333334
    removed_feature : PctExtNullSelfRedirectHyperlinksRT
    top_feature : PctExtNullSelfRedirectHyperlinksRT
    hyperparameter_regime : tuned_on_full_feature_set

```

[record 21 of 24]

```

        dataset : mendeley_tan
        iteration : 2
        n_features : 46
        accuracy : 0.9806666666666668
    fit_seconds : 2.32
    shap_seconds : 356.45
    mean_tree_depth : 20.0
    total_nodes : 310536
    window_start :
    window_end :
    cache_replay : False
    source_log : screening
    run_date :
    window_reconstructed :
        competing_jobs :
        contention_evidence : not_assessed
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
        sampled_job_tags :
        n_load_samples :
        contended :
    cost_analysis_eligible :
        remeasured_clean :
    shap_seconds_original_contended :
        f1 : 0.980653769179453
        mcc : 0.9613341878529912
    removed_feature : PctNullSelfRedirectHyperlinks
    top_feature : PctNullSelfRedirectHyperlinks
    hyperparameter_regime : tuned_on_full_feature_set

```

[record 22 of 24]

```

        dataset : mendeley_tan
        iteration : 3
        n_features : 45
        accuracy : 0.9753333333333334
    fit_seconds : 2.24
    shap_seconds : 372.43
    mean_tree_depth : 20.0
    total_nodes : 343486
    window_start :

```

**Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)**

```

        window_end :
        cache_replay : False
        source_log : screening
        run_date :
    window_reconstructed :
        competing_jobs :
        contention_evidence : not_assessed
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
        sampled_job_tags :
        n_load_samples :
        contended :
    cost_analysis_eligible :
        remeasured_clean :
    shap_seconds_original_contended :
        f1 : 0.9752839011356046
        mcc : 0.9506742720912652
    removed_feature : FrequentDomainNameMismatch
    top_feature : FrequentDomainNameMismatch
    hyperparameter_regime : tuned_on_full_feature_set

```

[record 23 of 24]

```

        dataset : mendeley_tan
        iteration : 4
        n_features : 44
        accuracy : 0.972
        fit_seconds : 2.22
        shap_seconds : 381.82
    mean_tree_depth : 20.0
    total_nodes : 382488
    window_start :
    window_end :
    cache_replay : False
    source_log : screening
    run_date :
    window_reconstructed :
        competing_jobs :
        contention_evidence : not_assessed
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
        sampled_job_tags :
        n_load_samples :
        contended :
    cost_analysis_eligible :
        remeasured_clean :
    shap_seconds_original_contended :
        f1 : 0.9720372836218376
        mcc : 0.9440033564623456
    removed_feature : PctExtResourceUrls
    top_feature : PctExtResourceUrls
    hyperparameter_regime : tuned_on_full_feature_set

```

[record 24 of 24]

```

        dataset : mendeley_tan
        iteration : 5
        n_features : 43
        accuracy : 0.9646666666666668
        fit_seconds : 2.19
        shap_seconds : 414.92
    mean_tree_depth : 20.0

```

**Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)**

```

total_nodes : 423066
window_start :
window_end :
cache_replay : False
source_log : screening
run_date :
window_reconstructed :
competing_jobs :
contention_evidence : not_assessed
sampled_cpu_percent_mean :
sampled_cpu_percent_max :
sampled_n_python_procs_max :
sampled_job_tags :
n_load_samples :
contended :
cost_analysis_eligible :
remeasured_clean :
shap_seconds_original_contended :
f1 : 0.9650165016501652
mcc : 0.9295192557785932
removed_feature :
top_feature : ExtMetaScriptLinkRT
hyperparameter_regime : tuned_on_full_feature_set

```

**Table T13. Directional agreement per shared concept, ranked by how much the disagreement matters.**

| concept               | rho_phiusiil        | rho_uci                | direction_agrees | abs_rho_phiusiil   | abs_rho_uci           | status        | importance_weight  | headline |
|-----------------------|---------------------|------------------------|------------------|--------------------|-----------------------|---------------|--------------------|----------|
| anchor_external_ratio | 0.0872798521005418  | -0.7012072591666815    | False            | 0.0872798521005418 | 0.7012072591666815    | INVERTED      | 0.7012072591666815 | True     |
| iframe                | 0.521331928092635   | 0.0033935238644643     | False            | 0.521331928092635  | 0.0033935238644643    | NO_SIGNAL_UCI | 0.521331928092635  | True     |
| ssl_state             | -0.6124154229775562 | -0.735813975062039     | True             | 0.6124154229775562 | 0.735813975062039     | agrees        | 0.735813975062039  | False    |
| url_length            | -0.3539937654486287 | -0.0533315794339325    | True             | 0.3539937654486287 | 0.0533315794339325    | agrees        | 0.3539937654486287 | False    |
| subdomain_depth       | -0.1141491829083289 | -0.3046922676521399    | True             | 0.1141491829083289 | 0.3046922676521399    | agrees        | 0.3046922676521399 | False    |
| form_handler          | 0.1673089014798561  | -0.2191900653063038    | False            | 0.1673089014798561 | 0.2191900653063038    | INVERTED      | 0.2191900653063038 | False    |
| popup                 | 0.1925460026806499  | -8.588679256217027e-05 | False            | 0.1925460026806499 | 8.588679256217027e-05 | NO_SIGNAL_UCI | 0.1925460026806499 | False    |
| ip_in_domain          | -0.0605489999124573 | -0.0941600949562038    | True             | 0.0605489999124573 | 0.0941600949562038    | agrees        | 0.0941600949562038 | False    |
| url_redirect          | 0.0480143528227923  | 0.0201134616575737     | True             | 0.0480143528227923 | 0.0201134616575737    | agrees        | 0.0480143528227923 | False    |

**Table T14. Concepts retained and lost when PhiUSIL is restricted to its leak-controlled feature set.**

| concept               | status                   | reason                                                                      | rho_uci                | rho_phiusiil        | polarity_status |
|-----------------------|--------------------------|-----------------------------------------------------------------------------|------------------------|---------------------|-----------------|
| ip_in_domain          | retained_after_deleaking |                                                                             | -0.0941600949562038    | -0.0605489999124573 | agrees          |
| url_length            | retained_after_deleaking |                                                                             | -0.0533315794339325    | -0.3539937654486287 | agrees          |
| subdomain_depth       | retained_after_deleaking |                                                                             | -0.3046922676521399    | -0.1141491829083289 | agrees          |
| iframe                | retained_after_deleaking |                                                                             | 0.0033935238644643     | 0.521331928092635   | NO_SIGNAL_UCI   |
| popup                 | retained_after_deleaking |                                                                             | -8.588679256217027e-05 | 0.1925460026806499  | NO_SIGNAL_UCI   |
| url_redirect          | retained_after_deleaking |                                                                             | 0.0201134616575737     | 0.0480143528227923  | agrees          |
| form_handler          | retained_after_deleaking |                                                                             | -0.2191900653063038    | 0.1673089014798561  | INVERTED        |
| ssl_state             | LOST_to_deleaking        | source feature(s) removed by de-leaking: ['IsHTTPS']                        | -0.735813975062039     | -0.6124154229775562 | agrees          |
| anchor_external_ratio | LOST_to_deleaking        | source feature(s) removed by de-leaking: ['NoOfExternalRef', 'NoOfSelfRef'] | -0.7012072591666815    | 0.0872798521005418  | INVERTED        |

**Table T16. Runtime observations for the reproducibility section, each labelled clean or contended.**

```
[record 1 of 1]
  observation : uci_full E2 explanation cost rose ~3.5x against the same data under the previous
                hyperparameters
measured_before_seconds : 447.0
measured_after_seconds : 1581.3
  ratio : 3.54
  driver : RandomForest n_estimators 100 -> 500
  driver_exact : True
  driver_detail : uci_full received its own hyperparameter search in A4 (justified: it carries 11,055 rows
                  against uci_dedup's 5,849). That search selected 500 trees where the earlier uci parameters
                  used 100 - a 5x increase in the forest the explainer must walk.
  mechanism : Interventional TreeSHAP cost scales with the total node count of the model being explained,
              so explanation cost tracks model complexity regardless of WHY the complexity changed.
  significance : An INDEPENDENT confirmation of the B5b mechanism through a different driver. In A1.3 node
                 count rose because features were removed and the boundary hardened; here it rose because a
                 hyperparameter search chose a larger forest. Same relationship, unrelated cause - which is
                 what makes explanation cost a general proxy for model complexity rather than an artifact of
                 the removal procedure.
before_timing_clean : False
after_timing_clean : False
  contention_note : NEITHER timing is uncontended: the 447s baseline ran alongside A1.3, and the 1581s figure
                  ran alongside table-building commands (load sampler shows job tags A4A5+B2 across
                  18:38-19:05). The 5x tree count is exact and structural; the ~3.5x ratio is approximate and
                  should be quoted as 'roughly 3.5x' or re-measured on an idle machine before being stated
                  precisely.
  n_explained : 5527
    source : A5 uci_full seed 42/43, half A
```